



NCIE-016

Testing, Verification & Acceptance Specification

**In development — for human review of full expansion —
Version 1.0**

Specifies the comprehensive verification programme for the approved NCIE-001–015 foundation: requirements traceability, test-level taxonomy, risk-based prioritization, environment/data control, automation architecture, code-level through cross-specification end-to-end testing, security/AI/domain/UX/operations/recovery/performance/resilience verification, defect governance, evidence/results model, execution independence, acceptance/sign-off, coverage reporting, reproducibility, governance-control verification, CI/CD continuous verification, and bounded production verification. This document specifies the verification programme only; it does not execute a single test, and nothing in it may be cited as an executed result, a pass, or an acceptance decision.

Verification-only, upstream-consuming: NCIE-016 owns detailed verification strategy, test-case design, the Requirements Traceability Matrix, execution governance, evidence and acceptance/sign-off matrices; it does not rewrite NCIE-001–015 requirements, fabricate implementation evidence, or claim any test executed or passed.

Expands approved NCIE-016 Skeletal v0.1, itself traced to approved NCIE-001 v1.3 through approved NCIE-015 v1.0.

Prepared: 16 September 2026

Maintained by: National Communications Intelligence Ecosystem (NCIE)

Document Governance

0.1 Formal-Title Reconciliation (Directive Item 3)

The approved NCIE-016 Skeletal v0.1 carries the working title 'NCIE-016 — Testing, Quality Assurance & Verification Specification' and its own Title-control note directs that the exact formal wording be reconciled against the controlled suite register before full expansion. The authoritative source located and used is the NCIE Documentation Suite Master Controlled Register v2.0 (FOR APPROVAL, dated 02 September 2026), Section 3, whose NCIE-016 entry records the formal title as 'Testing, Verification & Acceptance Specification' with Primary Purpose 'Verification strategy, test suites, security/AI/data testing and acceptance gates' — matching this document's actual 40-chapter scope (verification strategy Chapters 1-9, test suites across every upstream domain Chapters 10-28, and acceptance/sign-off governance Chapters 29-32) exactly. Comparison result: SEMANTIC MATCH — DOCUMENT-CONTROL WORDING DIFFERS (the register drops 'Quality Assurance' and adds 'Acceptance'; the underlying scope and Primary Purpose match precisely, so this is a wording correction only, per directive item 3, not a material scope discrepancy requiring Human Review halt). Per this result, the document-control formal title has been set to 'NCIE-016 — Testing, Verification & Acceptance Specification' throughout this document (cover, this section, Document Governance table, Authority Chain, running header, Chapter 40 disposition statement, core document properties and the deliverable filename); descriptive prose referring to the document's 'testing' or 'quality assurance' content throughout the chapters is unaffected, since that describes subject matter, not the formal document-control title. This determination is recorded as Human Review item HR16-1-1 for institutional confirmation rather than treated as silently final.

0.2 Authority Chain

NCIE-001 requirements → NCIE-002 architecture → NCIE-003 canonical data → NCIE-004 engineering baseline → NCIE-005 ARGUS behavioral authority → NCIE-006 Memory & Context architecture → NCIE-007 AI/Agent/Model Orchestration → NCIE-008 Governance, Authority, Accountability & RACI → NCIE-009 Security, IAM & Zero-Trust Architecture → NCIE-010 Evidence, Provenance & Audit Architecture → NCIE-011 API, Integration & Data Connector Specification → NCIE-012 UX/UI & Design System Specification → NCIE-013 Functional Modules Specification → NCIE-014 Database & Storage Engineering Specification → NCIE-015 DevSecOps, Infrastructure & Deployment Specification → NCIE-016 Testing, Verification & Acceptance Specification. NCIE-016 verifies governed requirements, architecture, security, data, domain, persistence and operational decisions; it creates no canonical meaning, institutional authority, security policy, Evidence semantics, API contract, UX pattern, domain-business rule, persistence semantic or operational procedure of its own — and, distinctively among this suite's specifications, it executes no test merely by describing one.

0.3 Token-Conservation Convention

Every chapter opens with a short Upstream block citing the specific upstream NCIE-001–015 authority the chapter verifies, stating only the verification relationship rather than restating upstream architectural, security, canonical, domain-business, persistence or operational content.

0.4 Governing Testing / Verification Invariants Register

These 21 invariants bind every chapter of this document and are stated once here rather than re-derived repeatedly. They are fixed by the approved NCIE-016 Skeletal v0.1 and by the approved NCIE-001–015 authority chain; no institutional decision is required to adopt them. The upstream canonical, governance, security, Evidence/provenance, integration, UX, domain-module, persistence and operational invariants already binding from NCIE-003 and NCIE-008–015 remain fully in force and are cross-referenced, not restated, in the chapters below.

- TEST SPECIFIED ≠ TEST EXECUTED.
- TEST EXECUTED ≠ TEST PASSED.
- TEST PASSED ≠ PRODUCTION ACCEPTANCE.
- IMPLEMENTATION EVIDENCE AVAILABLE ≠ TEST EXECUTION.
- AUTOMATED PASS ≠ HUMAN / INSTITUTIONAL ACCEPTANCE.
- UNIT PASS ≠ SYSTEM PASS.
- INTEGRATION PASS ≠ END-TO-END PASS.

- SCHEMA PASS ≠ SEMANTIC CORRECTNESS.
- SECURITY SCAN PASS ≠ SECURITY ASSURANCE.
- PENETRATION TEST PASS ≠ ZERO RISK.
- VPF PASS ≠ HUMAN APPROVAL.
- MODEL BENCHMARK PASS ≠ DOMAIN VALIDITY.
- AGENT TASK SUCCESS ≠ AUTHORITY COMPLIANCE.
- ACCESSIBILITY TOOL PASS ≠ ACCESSIBILITY CONFORMANCE.
- USABILITY SESSION ≠ REPRESENTATIVE-USER ACCEPTANCE.
- BACKUP TEST ≠ RECOVERY VALIDATION.
- DR EXERCISE ≠ PRODUCTION RECOVERY PROOF.
- PERFORMANCE TEST ≠ CAPACITY ASSURANCE.
- NO DEFECT OBSERVED ≠ DEFECT ABSENT.
- TRACEABILITY COMPLETE ≠ REQUIREMENT CORRECT.
- TEST DATA ≠ PRODUCTION AUTHORITY.
- DEFECT CLOSED ≠ ROOT CAUSE ELIMINATED.

0.5 Verification/Upstream Boundary Note

NCIE-016 owns detailed verification strategy, test-case design, the Requirements Traceability Matrix, execution governance, evidence and acceptance/sign-off matrices, and cross-specification quality validation. It consumes NCIE-001–015 as the sole substantive basis for what is verified; it does not redefine any of them, and it does not invent acceptance/sign-off authority, risk acceptance, test independence, representative users, or silently resolve Human Review. Where a matter requires a tool/product selection or a numeric threshold (coverage target, pass threshold, benchmark cutoff, performance limit, defect severity, RPO/RTO acceptance, flake tolerance, release-blocking threshold, test cadence) not fixed by an approved source, this document states the verification framework and defers the choice to Human Review rather than inventing an answer. Approval of the NCIE-016 Skeletal v0.1 authorized this full expansion only; it did not itself approve any tool selection, numeric threshold, acceptance authority or scenario-corpus assignment proposed below — and it did not execute a single test.

0.6 Human Review Focus Classification

Every chapter's Human Review Focus item carries one of seven labels: RESOLVED BY UPSTREAM BASELINE; PROPOSED DESIGN DEFAULT; INSTITUTIONAL CONFIRMATION REQUIRED; SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED; LEGAL/POLICY CONFIRMATION REQUIRED; ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED; or SOURCE/WORKFLOW DISCOVERY REQUIRED — each additionally marked Blocking or Non-Blocking, and each stating precisely what it blocks. Per directive item 70, a Chapter 39 register item that restates an identical NCIE-008–015 authority/threshold/environment/security/UX/operations question cross-references that upstream Human Review ID instead of duplicating it; a new NCIE-016 item is created only for a distinct testing decision. Chapter 39 consolidates every open item, extracted directly from the chapters so the register cannot drift from its sources (mechanically verified per §39.4).

0.7 Status and No-Execution-Claims Convention

Skeletal-structure approval authorizes expansion; it does not approve any individual tool selection, numeric threshold, acceptance authority or scenario corpus proposed below — those remain IN DEVELOPMENT / FOR HUMAN REVIEW until a separate approval act. Per directive items 20 and 82, specification prose, sample test cases, matrices or planned automation described anywhere in this document must never be represented as implementation evidence, execution, PASS, validation or acceptance; no test in this document has actually been run, and this document does not claim otherwise anywhere. This convention is restated in every chapter's status line so it cannot be overlooked.

0.8 Salient Things to Note Discipline

Directive item 86 requires this expansion to programmatically verify an explicit '<N>.0 Salient Things to Note' subsection in all 40 chapters, continuing the discipline established after the NCIE-009 Chapter-27 omission (found by independent human examination and corrected) and reinforced in every full expansion since. Every chapter below renders its Salient Things to Note subsection as an unconditional part of the same build-time helper (NCIE-016's lib.py raises a build error if a chapter's salient content is missing), and Chapter 40's mechanical audit (§40.2) re-verifies this against the final document.

0.9 Document Amendment History

Version: v0.1

Change: Approved skeletal review edition (structural/conceptual review only; full expansion not yet authorized).

Version: v1.0

Change: Full expansion of the approved NCIE-016 Skeletal v0.1 into the complete 40-chapter specification per the NCIE-016 v0.1→v1.0 Claude Full-Expansion Production Directive: six consolidated matrices (Master RTM, Test-State Matrix, Cross-Spec Test Catalogue, Defect-State Matrix, Test-Evidence Matrix, Acceptance/Sign-Off Matrix), status-integrity and no-execution-claims audits, Human Review Register (Chapter 39) and full traceability/handover (Chapter 40). Formal title reconciled against the NCIE Documentation Suite Master Controlled Register v2.0 at production time (\$0.1) — document-control wording corrected to 'Testing, Verification & Acceptance Specification'; no material scope discrepancy found.

Authority: Approved NCIE-001 Master PRD v1.3 through approved NCIE-015 DevSecOps, Infrastructure & Deployment Specification v1.0, and the approved NCIE-016 Skeletal v0.1.

Scope: Controlled full expansion only — the approved 40-chapter skeletal structure, chapter purposes and boundaries are preserved exactly; no chapter was added, removed, merged, split or renumbered. This expansion authorizes specification only; it does not authorize or imply test execution.

Status: v1.0 IN DEVELOPMENT — FOR HUMAN REVIEW OF FULL EXPANSION. The approved v0.1 skeleton remains historically preserved as the expansion architecture.

Document	NCIE-016 Testing, Verification & Acceptance Specification
Edition	In development — Version 1.0 (Full Expansion of Approved Skeletal v0.1)
Requirements authority	NCIE-001 Master Product Requirements Document, Version 1.3 — APPROVED
Architecture authority	NCIE-002 System Architecture & Technical Design, Version 0.5 — APPROVED
Canonical data authority	NCIE-003 Data Model & Data Dictionary, Version 0.4 — APPROVED
Engineering baseline	NCIE-004 Technology Stack & Engineering Blueprint, Version 1.2 — APPROVED
ARGUS behavioral authority	NCIE-005 ARGUS Intelligence Assistant Specification, Version 1.3 — APPROVED
Memory & Context authority	NCIE-006 Memory & Context Architecture Specification, Version 1.2 — APPROVED
Orchestration authority	NCIE-007 AI, Agent & Model Orchestration Specification, Version 1.2 — APPROVED
Governance authority	NCIE-008 Governance, Authority, Accountability & RACI Specification, Version 1.0 — APPROVED
Security authority	NCIE-009 Security, IAM & Zero-Trust Architecture, Version 1.0 — APPROVED
Evidence/provenance/audit authority	NCIE-010 Evidence, Provenance & Audit Architecture, Version 1.0 — APPROVED
Integration authority	NCIE-011 API, Integration & Data Connector Specification, Version 1.0 — APPROVED
UX/UI authority	NCIE-012 UX/UI & Design System Specification, Version 1.0 — APPROVED

Domain-module authority	NCIE-013 Functional Modules Specification, Version 1.0 — APPROVED
Persistence authority	NCIE-014 Database & Storage Engineering Specification, Version 1.0 — APPROVED
Operations authority	NCIE-015 DevSecOps, Infrastructure & Deployment Specification, Version 1.0 — APPROVED
Prior baseline	NCIE-016 Testing, Quality Assurance & Verification Specification (working title), Skeletal v0.1 — APPROVED FOR FULL EXPANSION (historically preserved)
Status	IN DEVELOPMENT — FOR HUMAN REVIEW OF FULL EXPANSION until explicitly approved; the approved v0.1 skeleton remains historically preserved as the expansion architecture. No test in this document has been executed.

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Human Review Focus 65

Mandate, Authority Chain & Verification Boundary

Establish NCIE-016 ownership of comprehensive testing/verification within the approved NCIE suite.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Specification only; no test in this document is executed, passed, or accepted by virtue of being specified here.

Upstream: NCIE-001–015.

1.0 Salient Things to Note

NCIE-016 verifies approved requirements; it cannot rewrite them or manufacture acceptance — Table 1 fixes this boundary and Table 2 maps every upstream domain to its required test classes without itself executing any of them.

1.1 Mandate and Scope

NCIE-016 verifies approved requirements, architecture, security, data, domain, persistence and operational decisions across NCIE-001–015 — it cannot rewrite them or manufacture acceptance. Nothing in this document proves that any requirement is met; it specifies how that would be verified.

1.2 Verification Boundary — What NCIE-016 Does Not Own

NCIE-016 does not own product requirements, architecture, canonical data, ARGUS behavior, Memory/Context semantics, Agent/VPF semantics, institutional governance, security policy, Evidence/provenance meaning, API contracts, UX semantics, domain business rules, persistence semantics or operational procedures. It owns only the verification layer above all of them.

Table 1. NCIE-016 verification boundary.

Concern	Owned By
Product requirements/scope	NCIE-001
System architecture	NCIE-002
Canonical data model	NCIE-003
Engineering baseline/technology families	NCIE-004
ARGUS behavioral authority	NCIE-005
Memory & Context architecture	NCIE-006
AI/Agent/Model orchestration	NCIE-007
Institutional governance, RACI	NCIE-008
Security, IAM, Zero-Trust enforcement	NCIE-009
Evidence lifecycle, provenance, audit	NCIE-010
API/event/connector/integration contracts	NCIE-011
UX/UI, presentation, interaction, Design System	NCIE-012
Domain business rules, workflows, analytical products	NCIE-013
Physical database/storage engineering	NCIE-014
Deployment/infrastructure realization, operations	NCIE-015
Verification strategy, RTM, evidence, acceptance	NCIE-016 (this document)

1.3 Cross-Spec Test Catalogue (directive item 65)

The consolidated catalogue below maps each NCIE-001–015 domain to its required test classes, owner/authority reference, evidence expectation and unresolved decision, without itself executing any of the listed tests.

Table 2. Cross-Spec Test Catalogue.

Domain	Required Test Classes	Owner/Authority Reference	Evidence Expectation	Unresolved Decision
Requirements/architecture (NCIE-001/002)	Traceability, regression, change-impact (Ch.3-4, 28)	NCIE-001/002 owners	RTM entries, regression run records	Baseline-freeze policy — HR16-3-1
Canonical data/persistence (NCIE-003/014)	Data-model, migration, reconciliation, corruption (Ch.12)	NCIE-003/014 owners	Migration/reconciliation reports	Migration acceptance authority — HR16-12-1
Memory/Context (NCIE-006)	Isolation, authz, search/vector, Decision-Time (Ch.13)	NCIE-006 owner	Scenario/adversarial test records	Memory scenario corpus — HR16-13-1
ARGUS (NCIE-005)	Conversational/behavioral scenarios (Ch.14)	NCIE-005 owner	Behavioral scenario transcripts	Behavioral corpus — HR16-14-1
AI/Agent/Model/Tool/VPF (NCIE-007)	Agent Factory, model evaluation, Tool/VPF testing (Ch.15-18)	NCIE-007 owner	Evaluation reports, VPF test cases	Benchmarks/thresholds — HR16-16-1
Security/IAM (NCIE-009)	Authn/authz, penetration, supply-chain (Ch.19)	NCIE-009 owner	Security-test reports	Security-test independence — HR16-19-1
Evidence/provenance/audit (NCIE-010)	Custody, tamper, Decision-Time, audit (Ch.20)	NCIE-010 owner	Defensibility test records	Defensibility acceptance — HR16-20-1
Domain modules (NCIE-013)	Rule/KPI/anomaly/cross-domain scenarios (Ch.21)	NCIE-013 owner	Domain scenario/validator records	Domain validators — HR16-21-1
UX/UI (NCIE-012)	Usability, accessibility, representative-user (Ch.22)	NCIE-012 owner	Accessibility/usability session records	Accessibility standard/level — HR16-22-1
Operations/DevSecOps (NCIE-015)	Pipeline, deployment, backup/DR, performance, resilience (Ch.23-26)	NCIE-015 owner	Operational test/exercise records	Workload/recovery targets — HR16-25-1

1.R RACI / Control Ownership

Table 3. Chapter 1 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Verification)	Consulted	Informed
Adopt/confirm NCIE-016 verification architecture	Accountable Authority Class (NCIE-008, to be assigned)	Verification Architecture Owner (to be assigned)	NCIE-001-015 chapter owners	All institutional role holders

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR16-1-1: Confirm the NCIE-016 formal title reconciliation (\$0.1 — corrected to 'Testing, Verification & Acceptance Specification' against Master Controlled Register v2.0), and verification/sign-off authority (skeletal Preliminary Human Review Theme 1).

Traceability: NCIE-001–015.

Quality Engineering & Verification Doctrine

Define common QA/verification principles governing every chapter this document specifies.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Specification only; no test in this document is executed, passed, or accepted by virtue of being specified here.

Upstream: NCIE-001–015.

2.0 Salient Things to Note

Verification, validation and acceptance are distinct and evidence-backed — Table 4 fixes seven non-substitutable states so no chapter can present one as if it were a later one.

2.1 Verification, Validation and Acceptance Are Distinct

Verification (does the implementation match the specification), validation (does it meet the actual need) and acceptance (an institutional decision to proceed) are three distinct, evidence-backed activities — this document repeatedly distinguishes them because collapsing any two is the single most common way a testing programme silently overclaims.

2.2 Risk-Based QA, Independence, Reproducibility and Transparency

Verification depth is risk-based (Chapter 6); execution independence scales with risk (Chapter 31); results must be reproducible (Chapter 34) and evidence-backed (Chapter 30); and residual gaps are reported transparently (Chapter 33) rather than hidden behind an aggregate pass rate.

2.3 Test-State Matrix (directive item 64)

The consolidated matrix below fixes an explicit state model — SPECIFIED, IMPLEMENTATION EVIDENCE AVAILABLE, EXECUTED, PASSED/FAILED/BLOCKED/NOT RUN, VALIDATED (where applicable), and HUMAN/PRODUCTION ACCEPTANCE — used consistently by every chapter below so no state is ever presented as a later one.

Table 4. Test-State Matrix.

State	Meaning	Who/What Establishes It	Never Implies
Specified	A test case/scenario is designed and documented in this specification	This document (Claude, under this directive)	That the test has ever been run
Implementation evidence available	Supporting implementation (code, config, harness) exists that could execute the test	Engineering delivery, if and when supplied	That the test has been executed against that implementation
Executed	The test was actually run against a specific environment/build	A named executor (Chapter 31), with a run record (Chapter 30)	That the test passed
Passed / Failed / Blocked / Not Run	The recorded outcome of an executed attempt (or the reason none occurred)	The executor/automation, reviewed per Chapter 31	Passed: that the broader capability is production-ready; Failed/Blocked: root cause (Ch.29)
Validated (where applicable)	A higher-order confirmation — e.g. recovery, migration, game-day — that a class of failure was survived under exercised	Domain-specific reviewer per the relevant chapter (e.g. Ch.24 for recovery)	Proof against every possible failure mode, or production-scale confirmation

	conditions		
Human/production acceptance	An institutional decision to accept the capability, taken with full visibility of results, defects and residual risk	NCIE-008 accountable authority (Chapter 32)	N/A — this is the terminal state

2.R RACI / Control Ownership

Table 5. Chapter 2 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Verification)	Consulted	Informed
Approve QA doctrine/independence	Accountable Authority Class (to be assigned)	QA Doctrine Owner (to be assigned)	NCIE-008 owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR16-2-1: Approve QA doctrine/independence (Table 4 state model).

Traceability: NCIE-001–015.

Requirements Baseline & Testable Acceptance Extraction

Turn approved requirements into testable assertions without guessing ambiguity.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Specification only; no test in this document is executed, passed, or accepted by virtue of being specified here.

Upstream: NCIE-001–015.

3.0 Salient Things to Note

TRACEABILITY COMPLETE ≠ REQUIREMENT CORRECT; unresolved requirements must be flagged — ambiguity is escalated as Human Review, never resolved by guessing.

3.1 Requirement IDs, Acceptance Statements and Source/Version

Every testable assertion in this document traces to a requirement ID, its source document and version; TRACEABILITY COMPLETE ≠ REQUIREMENT CORRECT — a fully traced requirement can still itself be wrong, ambiguous, or superseded upstream.

3.2 Testability, Ambiguity and HR/Change-Impact Handling

Where an approved requirement is not testable as written, or is ambiguous, this is flagged as an explicit Human Review item rather than resolved by guessing the intended meaning; a later upstream change is traced through the RTM (Chapter 4) as a change-impact event, not a silent baseline drift.

3.R RACI / Control Ownership

Table 6. Chapter 3 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Verification)	Consulted	Informed
Confirm baseline-freeze/ambiguity escalation	Accountable Authority Class (to be assigned)	Requirements Baseline Owner (to be assigned)	NCIE-001-015 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR16-3-1: Confirm baseline-freeze/ambiguity escalation policy.

Traceability: NCIE-001–015.

Requirements Traceability Matrix (RTM) Architecture

Define the full cross-suite RTM.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Specification only; no test in this document is executed, passed, or accepted by virtue of being specified here.

Upstream: NCIE-004 comprehensive verification programme.

4.0 Salient Things to Note

RTM linkage does not itself prove requirement correctness or test sufficiency — Table 7's Result column reads NOT RUN throughout because no test has actually been executed at the specification stage.

4.1 RTM Fields and Lifecycle

The RTM links requirement → test → environment → evidence → result → defect → acceptance as one continuous chain per requirement; RTM linkage does not itself prove requirement correctness or test sufficiency — it proves that a stated verification path exists and can be inspected.

4.2 Master RTM (directive item 63)

The excerpt below illustrates the Master RTM's structure across representative NCIE-001–015 requirement classes; the full RTM is populated per-requirement as each upstream specification is verified, and no result cell below is populated with a fabricated outcome — every Result cell in this specification-stage RTM reads NOT RUN.

Table 7. Master RTM (illustrative structure, no results fabricated).

Requirement ID	Source/Version	Acceptance Statement	Test ID(s)	Test Level	Environment	Evidence Ref	Result	Defect Ref	HR Dependency	Acceptance Authority/Status
NCIE-001-REQ-*	NCIE-001 v1.3	Per NCIE-001 requirement text	TBD per Ch.5 taxonomy	System/Acceptance	TBD (Ch.7)	TBD (Ch.30)	NOT RUN	N/A	HR16-3-1	Pending
NCIE-003-CANON-*	NCIE-003 v0.4	Canonical mapping/constraint per NCIE-003	TBD per Ch.12	Integration/System	TBD (Ch.7)	TBD (Ch.30)	NOT RUN	N/A	HR16-12-1	Pending
NCIE-009-SEC-*	NCIE-009 v1.0	Security control per NCIE-009	TBD per Ch.19	Security/Penetration	TBD (Ch.7)	TBD (Ch.30)	NOT RUN	N/A	HR16-19-1	Pending
NCIE-013-DOM-*	NCIE-013 v1.0	Domain rule/KPI per NCIE-013	TBD per Ch.21	System/Scenario	TBD (Ch.7)	TBD (Ch.30)	NOT RUN	N/A	HR16-21-1	Pending
NCIE-015-OPS-*	NCIE-015 v1.0	Operational control per NCIE-015	TBD per Ch.23-24	Operational/Recovery	TBD (Ch.7)	TBD (Ch.30)	NOT RUN	N/A	HR16-25-1	Pending

4.R RACI / Control Ownership

Table 8. Chapter 4 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Verification)	Consulted	Informed
Confirm RTM ownership/acceptance	Accountable Authority Class (to be assigned)	RTM Owner (to be assigned)	NCIE-001-015 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR16-4-1: Confirm RTM ownership/acceptance (Table 7 structure).

Traceability: NCIE-004 comprehensive verification programme.

Test Levels & Verification Taxonomy

Define verification levels from static/unit through institutional acceptance.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Specification only; no test in this document is executed, passed, or accepted by virtue of being specified here.

Upstream: NCIE-001–015.

5.0 Salient Things to Note

UNIT PASS ≠ SYSTEM PASS; INTEGRATION PASS ≠ END-TO-END PASS — each verification level is necessary but insufficient alone.

5.1 Static Through System Levels

Static analysis, unit, component, contract, integration and system levels each verify a different scope; UNIT PASS ≠ SYSTEM PASS — passing every unit test says nothing about whether the assembled system behaves correctly end to end.

5.2 E2E and Acceptance, and Non-Equivalence

End-to-end (E2E) and acceptance levels verify complete user/institutional journeys; INTEGRATION PASS ≠ END-TO-END PASS — two components integrating correctly pairwise does not prove the full chain behaves correctly. Each level is necessary but insufficient alone.

5.R RACI / Control Ownership

Table 9. Chapter 5 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Verification)	Consulted	Informed
Approve levels by risk	Accountable Authority Class (to be assigned)	Test Taxonomy Owner (to be assigned)	NCIE-001-015 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR16-5-1: Approve levels by risk.

Traceability: NCIE-001–015.

Risk-Based Test Prioritization

Prioritize verification by consequence/exposure without creating risk acceptance.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Specification only; no test in this document is executed, passed, or accepted by virtue of being specified here.

Upstream: NCIE-001–015.

6.0 Salient Things to Note

Test priority ≠ institutional risk acceptance — a capability given lower test depth carries an explicit residual-gap statement, never an implied full-coverage status.

6.1 Criticality, Impact and Exposure

Test priority and depth are driven by criticality, impact and exposure (e.g. security, financial, regulatory, safety consequence); test priority ≠ institutional risk acceptance — prioritizing a test highly does not itself accept the risk of not yet having run it.

6.2 Residual Gaps and Escalation

Any capability deliberately given lower test depth carries an explicit residual-gap statement (Chapter 33) rather than an implied 'fully covered' status; escalation to Human Review occurs when a residual gap touches a Blocking classification elsewhere in this document.

6.R RACI / Control Ownership

Table 10. Chapter 6 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Verification)	Consulted	Informed
Confirm prioritization/risk acceptance	Accountable Authority Class (to be assigned)	Risk-Based Prioritization Owner (to be assigned)	NCIE-001-015 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR16-6-1: Confirm prioritization/risk acceptance.

Traceability: NCIE-001–015.

Test Environment & Configuration Control

Control test environments/configurations for reproducibility.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Specification only; no test in this document is executed, passed, or accepted by virtue of being specified here.

Upstream: NCIE-001–015.

7.0 Salient Things to Note

Lower-environment pass ≠ production proof; record exact environment/configuration — equivalence to production is evidenced, never merely asserted.

7.1 Environment Identity and Configuration Record

Every test run records its exact environment identity, versions, configuration, dependencies, data set and identity/network/time context; lower-environment pass ≠ production proof — a test passing in a lower environment establishes what that environment demonstrated, nothing about production.

7.2 Reproducibility and Equivalence Evidence

Where a lower environment claims equivalence to production for verification purposes, that equivalence claim itself carries evidence (config diff, data-shape comparison) rather than being asserted; reproducibility is a property demonstrated per NCIE-015 Chapter 3's environment architecture, not assumed.

7.R RACI / Control Ownership

Table 11. Chapter 7 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Verification)	Consulted	Informed
Confirm environment equivalence	Accountable Authority Class (to be assigned)	Test Environment Owner (to be assigned)	NCIE-015 owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED] (Blocking) HR16-7-1: Confirm environment equivalence.

Traceability: NCIE-001–015.

Test Data, Synthetic Data & Privacy

Govern synthetic/masked/production-derived test data.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Specification only; no test in this document is executed, passed, or accepted by virtue of being specified here.

Upstream: NCIE-001–015.

8.0 Salient Things to Note

TEST DATA ≠ PRODUCTION AUTHORITY; protect sensitive/protected data — production-derived test data requires explicit Human Review authorization under NCIE-009 classification rules.

8.1 Synthetic, Masked and Production-Derived Data

Test data is synthetic or masked by default; any production-derived data requires explicit Human Review authorization under NCIE-009 classification rules, with minimization and access control equivalent to production.

8.2 Provenance, Refresh, Access and Cleanup

TEST DATA ≠ PRODUCTION AUTHORITY — a workflow succeeding against test data proves nothing about production authorization state. Test-data provenance, refresh cadence, access controls and post-use cleanup are all recorded so test data cannot silently persist beyond its authorized use.

8.R RACI / Control Ownership

Table 12. Chapter 8 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Verification)	Consulted	Informed
Confirm permitted production-derived data	Accountable Authority Class (to be assigned)	Test Data Owner (to be assigned)	NCIE-009 owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED] (Blocking) HR16-8-1: Confirm permitted production-derived data.

Traceability: NCIE-001–015.

Test Automation Architecture

Define automation framework requirements without unsupported product selection.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Specification only; no test in this document is executed, passed, or accepted by virtue of being specified here.

Upstream: NCIE-001–015.

9.0 Salient Things to Note

AUTOMATED PASS ≠ HUMAN / INSTITUTIONAL ACCEPTANCE — an automated green build is an execution signal, never itself an acceptance decision.

9.1 Harnesses, Runners, Fixtures and Mocks

Automation harness, runner, fixture, mock/stub and orchestration requirements are stated conceptually without mandating an unsourced product; AUTOMATED PASS ≠ HUMAN / INSTITUTIONAL ACCEPTANCE — an automated green build is an execution signal, not an acceptance decision.

9.2 Evidence Capture and Flakiness Handling

Automation captures evidence (Chapter 30) for every run regardless of outcome; flaky results are classified and quarantined per Chapter 34 rather than silently retried until green.

9.R RACI / Control Ownership

Table 13. Chapter 9 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Verification)	Consulted	Informed
Confirm automation technology/coverage goals	Accountable Authority Class (to be assigned)	Test Automation Owner (to be assigned)	NCIE-004 owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED] (Blocking) HR16-9-1: Confirm automation technology/coverage goals (no product mandated by this document).

Traceability: NCIE-001–015.

Static Analysis, Code Quality & Unit Verification

Define code-level/static/unit verification.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Specification only; no test in this document is executed, passed, or accepted by virtue of being specified here.

Upstream: NCIE-001–015.

10.0 Salient Things to Note

Static/unit success ≠ system/security/business correctness — a clean static or unit pass says nothing about the assembled system's security or business correctness.

10.1 Static Analysis, Lint, Type and Unit Concepts

Static analysis, lint, type-checking, unit and property/mutation-testing concepts are stated as a code-quality layer; static/unit success ≠ system/security/business correctness — a clean static pass says nothing about whether the assembled system is secure or correct.

10.2 Coverage Evidence and Code-Defect Handling

Coverage evidence is captured per the run record (Chapter 30); code-level defects follow the same defect-governance path (Chapter 29) as any other defect, never silently fixed outside the recorded process.

10.R RACI / Control Ownership

Table 14. Chapter 10 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Verification)	Consulted	Informed
Confirm quality gates/coverage targets	Accountable Authority Class (to be assigned)	Code Quality Owner (to be assigned)	NCIE-004 owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR16-10-1: Confirm quality gates/coverage targets (no numeric value invented by this document).

Traceability: NCIE-001–015.

API, Contract & Integration Testing

Execute detailed NCIE-011 contract/integration verification.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Specification only; no test in this document is executed, passed, or accepted by virtue of being specified here.

Upstream: NCIE-011 Ch.30.

11.0 Salient Things to Note

SCHEMA PASS ≠ SEMANTIC CORRECTNESS; test retry/idempotency/outcome unknown — a schema-valid payload can still carry the wrong business meaning.

11.1 Contract, Compatibility and Authorization

Contract and compatibility tests verify NCIE-011's API/event contracts hold across versions; SCHEMA PASS ≠ SEMANTIC CORRECTNESS — a schema-valid payload can still carry the wrong business meaning. Authorization tests confirm every endpoint enforces NCIE-009 policy.

11.2 Replay, Idempotency, Outcome-Unknown and Connector Faults

Replay/duplicate-suppression, retry/idempotency and outcome-unknown scenarios are tested explicitly, since these are the classes of defect that pass a naive happy-path test while failing under real network conditions; connector-fault injection and partner-certification scenarios are scoped per Chapter 26.

11.R RACI / Control Ownership

Table 15. Chapter 11 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Verification)	Consulted	Informed
Confirm partner/certification acceptance	Accountable Authority Class (to be assigned)	API/Integration Test Owner (to be assigned)	NCIE-011 owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR16-11-1: Confirm partner/certification acceptance.

Traceability: NCIE-011 Ch.30.

Data Model, Database & Migration Testing

Verify NCIE-003/014 data, persistence and migration behavior.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Specification only; no test in this document is executed, passed, or accepted by virtue of being specified here.

Upstream: NCIE-003; NCIE-014 Ch.32.

12.0 Salient Things to Note

Migration/CRUD success ≠ semantic reconciliation correctness — a technically clean migration can still misrepresent meaning if the underlying mapping was wrong.

12.1 Constraints, Mappings and Migration

Constraint, canonical-mapping and migration tests verify NCIE-003/014's physical realization matches canonical semantics; migration/CRUD success ≠ semantic reconciliation correctness — a technically clean migration can still misrepresent meaning if the mapping itself was wrong.

12.2 Backfill, Reconciliation and Corruption

Backfill and reconciliation tests verify NCIE-014 Chapter 26-27's discipline; corruption-detection tests confirm integrity mechanisms actually catch tampered/corrupted data rather than assuming they do.

12.R RACI / Control Ownership

Table 16. Chapter 12 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Verification)	Consulted	Informed
Confirm migration/reconciliation acceptance	Accountable Authority Class (to be assigned)	Data/Migration Test Owner (to be assigned)	NCIE-003/014 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR16-12-1: Confirm migration/reconciliation acceptance.

Traceability: NCIE-003; NCIE-014 Ch.32.

Memory, Context, Search & Retrieval Testing

Verify NCIE-006 Memory/Context/search/retrieval semantics.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Specification only; no test in this document is executed, passed, or accepted by virtue of being specified here.

Upstream: NCIE-006 Ch.34; NCIE-009.

13.0 Salient Things to Note

Retrieval relevance ≠ authorization/truth; test leakage and Decision-Time contamination — a highly relevant item must still pass its own authorization check before disclosure.

13.1 Isolation, Authorization and Vector/Search Relevance

Isolation and authorization tests confirm one user/Room/Investigation's Memory cannot leak into another's; retrieval relevance ≠ authorization/truth — a highly relevant retrieved item must still pass its own authorization check before disclosure.

13.2 Expiry, Contradiction and Decision-Time Contamination

Expiry/invalidation tests confirm stale Memory does not silently persist past its governed lifetime; adversarial tests specifically probe for contradiction handling and Decision-Time contamination — a later-corrected fact leaking into a Decision made before the correction.

13.R RACI / Control Ownership

Table 17. Chapter 13 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Verification)	Consulted	Informed
Confirm Memory scenarios/acceptance	Accountable Authority Class (to be assigned)	Memory Test Owner (to be assigned)	NCIE-006/009 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR16-13-1: Confirm Memory scenarios/acceptance.

Traceability: NCIE-006 Ch.34; NCIE-009.

ARGUS Behavioral & Conversational Testing

Verify NCIE-005 ARGUS behavior and degraded/Human-Primary responses.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Specification only; no test in this document is executed, passed, or accepted by virtue of being specified here.

Upstream: NCIE-005; NCIE-012.

14.0 Salient Things to Note

Helpful response ≠ compliant response; preserve uncertainty/Evidence/authority — a response that satisfies the user can still violate authority, privacy or Human-Primary boundaries.

14.1 Conversation, Domain Assistance and Rooms

Behavioral scenario testing covers conversation flow, Domain Assistance responses, Room context and voice interaction; helpful response ≠ compliant response — a response that satisfies the user's request can still violate authority, privacy or Human-Primary boundaries.

14.2 Personalization, Privacy and Degraded-Mode Behavior

Personalization and privacy scenarios confirm ARGUS preserves uncertainty, Evidence status and authority boundaries even when being maximally helpful; degraded-mode scenarios confirm graceful, non-deceptive behavior under upstream failure.

14.R RACI / Control Ownership

Table 18. Chapter 14 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Verification)	Consulted	Informed
Confirm behavioral scenario corpus	Accountable Authority Class (to be assigned)	ARGUS Test Owner (to be assigned)	NCIE-005/012 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR16-14-1: Confirm behavioral scenario corpus.

Traceability: NCIE-005; NCIE-012.

Agent Factory, Agent Runtime & Multi-Agent Testing

Verify NCIE-007 Agent Factory/runtime/multi-Agent behavior.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Specification only; no test in this document is executed, passed, or accepted by virtue of being specified here.

Upstream: NCIE-007 Ch.32; NCIE-009.

15.0 Salient Things to Note

AGENT TASK SUCCESS ≠ AUTHORITY COMPLIANCE — a task completing successfully says nothing about whether the Agent stayed within its authorized boundary while doing so.

15.1 Discovery, Synthesis and Gate A

Agent discovery/synthesis and Gate A approval-binding tests confirm an Agent cannot become operational without the governed approval NCIE-007 requires; AGENT TASK SUCCESS ≠ AUTHORITY COMPLIANCE — a task completing successfully says nothing about whether the Agent stayed within its authorized boundary while doing so.

15.2 Sandbox, Delegation, Multi-Agent and Cleanup

Sandbox isolation, delegation-chain and multi-Agent interaction tests confirm NCIE-015 Chapter 11's runtime boundaries hold under composition; cleanup tests confirm ephemeral resources are actually released, not merely scheduled for release.

15.R RACI / Control Ownership

Table 19. Chapter 15 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Verification)	Consulted	Informed
Confirm Agent corpus/generated-code scope	Accountable Authority Class (to be assigned)	Agent Test Owner (to be assigned)	NCIE-007/009 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED] (Blocking) HR16-15-1: Confirm Agent corpus/generated-code scope.

Traceability: NCIE-007 Ch.32; NCIE-009.

Model Gateway & AI Evaluation

Evaluate model routing/output robustness and domain fitness.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Specification only; no test in this document is executed, passed, or accepted by virtue of being specified here.

Upstream: NCIE-007/009/013.

16.0 Salient Things to Note

MODEL BENCHMARK PASS ≠ DOMAIN VALIDITY — a model scoring well on a general benchmark has not thereby been shown valid for this programme's specific domain use.

16.1 Routing, Eligibility, Fallback and Provenance

Model routing/eligibility/fallback tests confirm the Model Gateway selects and records provenance for the correct model per request; MODEL BENCHMARK PASS ≠ DOMAIN VALIDITY — a model scoring well on a general benchmark has not thereby been shown valid for this programme's specific domain use.

16.2 Quality, Robustness and Drift

Output-quality, robustness (adversarial-input resistance) and drift-detection evaluation are specified conceptually; benchmark sets and pass thresholds are Human Review decisions, never invented by this document.

16.R RACI / Control Ownership

Table 20. Chapter 16 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Verification)	Consulted	Informed
Confirm benchmarks/evaluation sets/thresholds	Accountable Authority Class (to be assigned)	AI Evaluation Owner (to be assigned)	NCIE-007/009/013 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR16-16-1: Confirm benchmarks/evaluation sets/thresholds (no value invented by this document).

Traceability: NCIE-007/009/013.

Tool Gateway & Consequential-Action Testing

Verify Tool/external consequential-action semantics.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Specification only; no test in this document is executed, passed, or accepted by virtue of being specified here.

Upstream: NCIE-007/011/009.

17.0 Salient Things to Note

Request accepted ≠ action completed; test stale approvals/retry duplication/outcome unknown — these are exactly the conditions under which a consequential action can double-execute or silently fail.

17.1 Authorization, Approval Binding and Idempotency

Authorization and approval-binding tests confirm a consequential Tool call cannot execute without its bound approval; request accepted ≠ action completed — a request being accepted for processing is not evidence the downstream consequential action actually occurred.

17.2 Timeout, Receipt, Outcome-Unknown and Compensation

Stale-approval, retry-duplication and outcome-unknown scenarios are tested explicitly, since these are exactly the conditions under which a consequential action can double-execute or silently fail; reconciliation and compensation/revocation paths are tested for every high-risk Tool scenario.

17.R RACI / Control Ownership

Table 21. Chapter 17 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Verification)	Consulted	Informed
Confirm high-risk Tool scenarios	Accountable Authority Class (to be assigned)	Tool Gateway Test Owner (to be assigned)	NCIE-007/009/011 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED] (Blocking) HR16-17-1: Confirm high-risk Tool scenarios.

Traceability: NCIE-007/011/009.

VPF Validation System Testing

Verify VPF profiles, dispositions, history and revalidation.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Specification only; no test in this document is executed, passed, or accepted by virtue of being specified here.

Upstream: NCIE-003/007/009.

18.0 Salient Things to Note

VPF PASS ≠ HUMAN APPROVAL; test every disposition/history state — a passing VPF disposition is an automated validation signal, never the institutional approval act.

18.1 Profiles, Gate A/B and Dispositions

VPF profile, Gate A/B and disposition tests confirm every governed validation state is reachable and correctly recorded; VPF PASS ≠ HUMAN APPROVAL — a passing VPF disposition is an automated validation signal, never itself the institutional approval act.

18.2 Revalidation, Invalidation and History

Revalidation and invalidation tests confirm a VPF's disposition correctly expires or updates when its underlying conditions change; history tests confirm every prior disposition remains queryable rather than being overwritten.

18.R RACI / Control Ownership

Table 22. Chapter 18 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Verification)	Consulted	Informed
Confirm VPF golden cases/authority	Accountable Authority Class (to be assigned)	VPF Test Owner (to be assigned)	NCIE-003/007/009 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR16-18-1: Confirm VPF golden cases/authority.

Traceability: NCIE-003/007/009.

Security, IAM & Zero-Trust Testing

Execute detailed NCIE-009 security verification.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Specification only; no test in this document is executed, passed, or accepted by virtue of being specified here.

Upstream: NCIE-009 Ch.26.

19.0 Salient Things to Note

SECURITY SCAN PASS ≠ SECURITY ASSURANCE; PENETRATION TEST PASS ≠ ZERO RISK — both demonstrate resistance to what was actually tried, never the absence of all vulnerabilities.

19.1 Authn/Authz, Delegation and Privilege

Authentication/authorization, delegation and privilege-escalation tests verify NCIE-009's Zero-Trust controls hold under adversarial conditions; SECURITY SCAN PASS ≠ SECURITY ASSURANCE — a clean automated scan detects known patterns, not the absence of all vulnerabilities.

19.2 Secrets, Network, DLP and Prompt Injection

Secrets handling, network segmentation, DLP and prompt-injection resistance are tested as distinct classes; PENETRATION TEST PASS ≠ ZERO RISK — a penetration test scoped to a timeframe and methodology demonstrates resistance to what was actually tried, not the absence of all possible attacks.

19.R RACI / Control Ownership

Table 23. Chapter 19 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Verification)	Consulted	Informed
Confirm security-test independence/acceptance	Accountable Authority Class (to be assigned)	Security Test Owner (to be assigned)	NCIE-009 owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED] (Blocking) HR16-19-1: Confirm security-test independence/acceptance.

Traceability: NCIE-009 Ch.26.

Evidence, Provenance, Audit & Defensibility Testing

Execute NCIE-010 Evidence/provenance/audit verification.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Specification only; no test in this document is executed, passed, or accepted by virtue of being specified here.

Upstream: NCIE-010 Ch.27.

20.0 Salient Things to Note

Integrity/immutability ≠ truth; demonstrate custody/contradiction/Decision-Time/audit behavior — a cryptographically intact record can still be factually wrong.

20.1 Custody, Tamper and Version Tests

Custody-chain, tamper-evidence and version-history tests confirm NCIE-010's integrity mechanisms actually detect what they claim to detect; integrity/immutability ≠ truth — a cryptographically intact record can still be factually wrong, and these tests do not confuse the two.

20.2 Contradiction, Decision-Time, Audit and Recovery

Contradiction-handling and Decision-Time tests confirm a later correction does not silently alter what an earlier Decision saw; audit-completeness and recovery tests confirm the audit trail survives the recovery scenarios defined in NCIE-014/015.

20.R RACI / Control Ownership

Table 24. Chapter 20 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Verification)	Consulted	Informed
Confirm defensibility evidence/acceptance	Accountable Authority Class (to be assigned)	Evidence Test Owner (to be assigned)	NCIE-010 owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR16-20-1: Confirm defensibility evidence/acceptance.

Traceability: NCIE-010 Ch.27.

Functional Module & Domain Scenario Validation

Execute NCIE-013 domain scenario validation.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Specification only; no test in this document is executed, passed, or accepted by virtue of being specified here.

Upstream: NCIE-013 Ch.32.

21.0 Salient Things to Note

ANOMALY ≠ FRAUD; KPI breach ≠ violation; tests cannot manufacture Findings — a passing scenario confirms computational correctness, never an underlying allegation's truth.

21.1 Rules, KPI and Anomaly Scenarios

Rule, KPI and anomaly scenario tests confirm NCIE-013's domain logic behaves as specified; ANOMALY ≠ FRAUD and KPI breach ≠ violation remain true throughout — these tests validate computational correctness, never institutional Findings.

21.2 Alerts, Cross-Domain, Evidence and Degraded Data

Alert-generation, cross-domain-correlation, domain-Evidence-interface and degraded-data scenarios are tested explicitly; tests cannot manufacture Findings — a scenario passing confirms the module computed correctly, not that any underlying allegation is true.

21.R RACI / Control Ownership

Table 25. Chapter 21 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Verification)	Consulted	Informed
Confirm domain scenarios/validators	Accountable Authority Class (to be assigned)	Domain Scenario Test Owner (to be assigned)	NCIE-013 owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR16-21-1: Confirm domain scenarios/validators.

Traceability: NCIE-013 Ch.32.

UX, Accessibility & Representative-User Validation

Execute NCIE-012 UX/accessibility/user validation.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Specification only; no test in this document is executed, passed, or accepted by virtue of being specified here.

Upstream: NCIE-012 Ch.35.

22.0 Salient Things to Note

ACCESSIBILITY TOOL PASS ≠ ACCESSIBILITY CONFORMANCE; usability session ≠ representative-user acceptance — an automated scan and an unrepresentative session both fall short of the real standard.

22.1 Usability, Accessibility and Keyboard/Screen-Reader

Usability, automated-accessibility, keyboard-navigation and screen-reader tests verify NCIE-012's design-system implementation; ACCESSIBILITY TOOL PASS ≠ ACCESSIBILITY CONFORMANCE — an automated scanner catches only a subset of conformance criteria.

22.2 Responsive, Security-State, Voice and Representative Users

Responsive-layout, security-state-presentation and voice-interaction tests are specified alongside representative-user validation; USABILITY SESSION ≠ REPRESENTATIVE-USER ACCEPTANCE — a usability session with any participants is not automatically representative of the actual user population without a deliberate sampling decision.

22.R RACI / Control Ownership

Table 26. Chapter 22 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Verification)	Consulted	Informed
Confirm accessibility level/user groups	Accountable Authority Class (to be assigned)	UX/Accessibility Test Owner (to be assigned)	NCIE-012 owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR16-22-1: Confirm accessibility level/user groups.

Traceability: NCIE-012 Ch.35.

DevSecOps, Deployment & Pipeline Testing

Execute NCIE-015 pipeline/deployment verification.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Specification only; no test in this document is executed, passed, or accepted by virtue of being specified here.

Upstream: NCIE-015 Ch.34.

23.0 Salient Things to Note

PIPELINE PASS ≠ PRODUCTION ACCEPTANCE; test rollback/emergency paths — paths exercised less often are correspondingly more likely to be broken.

23.1 Pipeline and Deployment

Pipeline-gate and deployment tests verify NCIE-015's CI/CD controls behave as specified; PIPELINE PASS ≠ PRODUCTION ACCEPTANCE remains true here exactly as it does in NCIE-015 itself — this chapter tests the pipeline, it does not thereby accept anything into production.

23.2 Rollback and Emergency Paths

Rollback, secret-rotation and drift-detection tests specifically probe the emergency and recovery paths NCIE-015 defines, since these are exercised far less often than the happy-path deployment and are correspondingly more likely to be broken.

23.R RACI / Control Ownership

Table 27. Chapter 23 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Verification)	Consulted	Informed
Confirm operational test acceptance	Accountable Authority Class (to be assigned)	Operations Test Owner (to be assigned)	NCIE-015 owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR16-23-1: Confirm operational test acceptance.

Traceability: NCIE-015 Ch.34.

Backup, Restore, DR & Game-Day Validation

Validate backup/restore/DR/game-day behavior.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Specification only; no test in this document is executed, passed, or accepted by virtue of being specified here.

Upstream: NCIE-014/015/009/010.

24.0 Salient Things to Note

BACKUP TEST ≠ RECOVERY VALIDATION; DR EXERCISE ≠ PRODUCTION RECOVERY PROOF — a readable backup file is not a validated recovery, and an exercise proves only the exercised scenario.

24.1 Restore, PITR and Failover

Restore, point-in-time recovery (PITR) and failover tests verify NCIE-014/015's recovery architecture; BACKUP TEST ≠ RECOVERY VALIDATION — confirming a backup file exists and is readable is not the same as confirming a full recovery actually restores correct, current, reconciled state.

24.2 Failback, Reconciliation and Game-Day

Failback and reconciliation tests confirm the round trip closes cleanly; DR EXERCISE ≠ PRODUCTION RECOVERY PROOF — a successful game-day exercise validates the exercised scenario under exercise conditions, not production recovery under real failure.

24.R RACI / Control Ownership

Table 28. Chapter 24 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Verification)	Consulted	Informed
Confirm recovery scenarios/RPO-RTO acceptance	Accountable Authority Class (to be assigned)	Recovery Test Owner (to be assigned)	NCIE-009/010/014/015 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR16-24-1: Confirm recovery scenarios/RPO-RTO acceptance.

Traceability: NCIE-014/015/009/010.

Performance, Load, Scalability & Capacity Testing

Verify performance/load/scalability/capacity.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Specification only; no test in this document is executed, passed, or accepted by virtue of being specified here.

Upstream: NCIE-002 NFRs; NCIE-014/015.

25.0 Salient Things to Note

PERFORMANCE TEST ≠ CAPACITY ASSURANCE; workload/environment must be explicit — a number without workload/environment context is never reported.

25.1 Load, Stress, Soak and Spike

Load, stress, soak and spike test classes are defined with an explicit workload and environment for every result — a number without its workload/environment context is meaningless and is never reported as such.

25.2 Concurrency, Latency, Saturation and the Capacity Boundary

PERFORMANCE TEST ≠ CAPACITY ASSURANCE — a performance test demonstrates behavior under the specific tested load; it does not prove the system has assured capacity for every future load. Concurrency, latency, throughput, saturation and scaling-degradation behavior are all tested as distinct dimensions.

25.R RACI / Control Ownership

Table 29. Chapter 25 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Verification)	Consulted	Informed
Confirm workload models/targets	Accountable Authority Class (to be assigned)	Performance Test Owner (to be assigned)	NCIE-002/014/015 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED] (Blocking) HR16-25-1: Confirm workload models/targets (no numeric value invented by this document).

Traceability: NCIE-002 NFRs; NCIE-014/015.

Resilience, Fault Injection & Chaos Validation

Verify controlled fault/failure resilience.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Specification only; no test in this document is executed, passed, or accepted by virtue of being specified here.

Upstream: NCIE-004 resilience; NCIE-015.

26.0 Salient Things to Note

One fault-survival scenario ≠ universal resilience; safety/abort boundaries are mandatory — surviving one injected fault says nothing about untested fault combinations.

26.1 Dependency, Network, Storage and Provider Faults

Controlled fault injection (dependency failure, network partition, storage fault, provider outage) is scoped to explicit abort criteria and safe environments; one fault-survival scenario ≠ universal resilience — surviving one injected fault says nothing about untested fault combinations.

26.2 Containment, Abort and Recovery

Safety and abort boundaries are mandatory for every fault-injection exercise; containment confirms the injected fault cannot escape its intended blast radius, and recovery confirms the system returns to a known-good state after the exercise concludes.

26.R RACI / Control Ownership

Table 30. Chapter 26 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Verification)	Consulted	Informed
Confirm fault-injection authority/safe environments	Accountable Authority Class (to be assigned)	Resilience Test Owner (to be assigned)	NCIE-004/015 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED] (Blocking) HR16-26-1: Confirm fault-injection authority/safe environments.

Traceability: NCIE-004 resilience; NCIE-015.

Observability, Alerting & Incident-Response Verification

Verify observability/alert/on-call/incident paths.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Specification only; no test in this document is executed, passed, or accepted by virtue of being specified here.

Upstream: NCIE-015 Ch.20–24; NCIE-010.

27.0 Salient Things to Note

Telemetry visibility ≠ root cause/incident correctness — seeing a signal is not the same as correctly diagnosing it.

27.1 Metrics/Logs/Traces and Alert Routing

Observability-path tests confirm metrics/logs/traces actually reach dashboards and alerts actually route to on-call per NCIE-015 Chapter 20-22; telemetry visibility ≠ root cause/incident correctness — seeing a signal is not the same as correctly diagnosing it.

27.2 On-Call, Incident, Runbooks and Evidence

End-to-end incident-response exercises test the full NCIE-015 Chapter 23-24 chain (detection through closure) including runbook execution and evidence capture, without asserting the exercised incident was a real one.

27.R RACI / Control Ownership

Table 31. Chapter 27 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Verification)	Consulted	Informed
Confirm incident exercise/acceptance	Accountable Authority Class (to be assigned)	Observability Test Owner (to be assigned)	NCIE-010/015 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR16-27-1: Confirm incident exercise/acceptance.

Traceability: NCIE-015 Ch.20–24; NCIE-010.

Regression, Compatibility & Change-Impact Testing

Verify regression and compatibility after change.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Specification only; no test in this document is executed, passed, or accepted by virtue of being specified here.

Upstream: NCIE-001–015.

28.0 Salient Things to Note

Local change pass ≠ suite compatibility; trace cross-spec impact — regression scope is derived from actual dependency structure, never guessed.

28.1 Regression and Cross-Version Compatibility

Regression tests re-verify previously passing behavior after a change; local change pass ≠ suite compatibility — a change passing its own local tests can still break a downstream consumer's API, schema, config, model, data or UX assumption.

28.2 Cross-Spec Impact Tracing

Change-impact tracing follows the RTM (Chapter 4) to identify every downstream NCIE specification a change could affect, so regression scope is derived from actual dependency structure rather than guessed.

28.R RACI / Control Ownership

Table 32. Chapter 28 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Verification)	Consulted	Informed
Confirm regression/release blocking	Accountable Authority Class (to be assigned)	Regression Test Owner (to be assigned)	NCIE-001-015 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR16-28-1: Confirm regression/release blocking policy.

Traceability: NCIE-001–015.

Defect, Triage, Root-Cause & Corrective-Action Governance

Define defect/RCA/corrective-action governance.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Specification only; no test in this document is executed, passed, or accepted by virtue of being specified here.

Upstream: NCIE-001–015.

29.0 Salient Things to Note

DEFECT CLOSED ≠ ROOT CAUSE ELIMINATED — Table 33 gives deferred/accepted its own governed closure path distinct from root-cause elimination, so closure records never look identical regardless of which actually happened.

29.1 Defect ID, Severity, Reproduction and Triage

Every defect carries a stable ID, severity, reproduction steps and evidence; triage assigns an owner and priority per approved policy, never left to whoever noticed it first.

29.2 Fix, Retest, RCA and the Closure Boundary

DEFECT CLOSED ≠ ROOT CAUSE ELIMINATED — a defect can be closed because its symptom was fixed, because it was deferred/accepted under governed risk acceptance, or because root cause was actually eliminated; closure records which of these applies rather than presenting all closures identically.

29.3 Defect-State Matrix (directive item 68)

The consolidated matrix below defines defect states and transitions, including deferred/accepted-where-governed as a distinct closure path from root-cause elimination.

Table 33. Defect-State Matrix.

State	Meaning	Transition In	Transition Out	Authority
Discovered	Defect first recorded with evidence	Test execution or observation	→ Triage	Executor/observer (Ch.31)
Triage	Severity/priority/owner assigned	← Discovered	→ Assigned	Triage owner
Assigned	Owner accepted responsibility for fix	← Triage	→ Fixed or → Deferred/Accepted	Assigned owner
Fixed	A candidate fix has been applied	← Assigned	→ Retest Pending	Fix implementer
Retest pending	Awaiting re-execution of the failing test	← Fixed	→ Verified or → Reopened	Executor (independent per Ch.31 where risk requires)
Verified	Retest confirms the symptom no longer reproduces	← Retest Pending	→ Closed	Reviewer (Ch.31)
Deferred/accepted (where governed)	Institutional decision to accept the residual risk without a fix	← Assigned or Triage	→ Closed (with residual-risk record)	NCIE-008 accountable authority — never the fix implementer alone
Reopened	A closed or verified defect recurs	← Verified or Closed	→ Triage	Any observer, with evidence
Closed	Terminal state for this defect instance	← Verified or Deferred/Accepted	N/A unless reopened	Reviewer/accountable authority per closure

path

29.R RACI / Control Ownership

Table 34. Chapter 29 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Verification)	Consulted	Informed
Confirm defect severity/closure authority	Accountable Authority Class (to be assigned)	Defect Governance Owner (to be assigned)	NCIE-001-015 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR16-29-1: Confirm defect severity/closure authority (Table 33).

Traceability: NCIE-001–015.

Test Evidence, Results & Verification Artifact Model

Define attributable/versioned test evidence/results.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Specification only; no test in this document is executed, passed, or accepted by virtue of being specified here.

Upstream: NCIE-001–015.

30.0 Salient Things to Note

Screenshots/logs alone are insufficient unless attributable, versioned and requirement-linked — Table 35 fixes the minimum evidence fields required per test class.

30.1 Test Case, Run and Environment Identity

Every test run records its test-case identity, run ID, environment (Chapter 7), inputs, outputs and artifacts (logs, screenshots) together, never any one of these in isolation; screenshots/logs alone are insufficient unless attributable, versioned and requirement-linked.

30.2 Result, Reviewer and Provenance

Every recorded result carries a reviewer and provenance chain back to the source versions tested, aligned to NCIE-010's Evidence model so a test-evidence record can itself withstand the same defensibility scrutiny as any other Evidence object.

30.3 Test-Evidence Matrix (directive item 67)

The consolidated matrix below defines the minimum evidence required by test class.

Table 35. Test-Evidence Matrix.

Test Class	Run Ident ity	Environmen t/Config	Input	Outp ut	Logs/Art ifacts	Timest amps	Actor/ Tool	Sour ce Versi ons	Resu lt	Review er
Unit/component (Ch.10)	Required	Required	Required	Required	Required	Required	Required (tool)	Required	Required	Optional (risk-based)
Integration/contract (Ch.11)	Required	Required	Required	Required	Required	Required	Required	Required	Required	Required
Security/penetration (Ch.19)	Required	Required	Required	Required	Required	Required	Required (named individual)	Required	Required	Required (independent)
Performance/resilience (Ch.25-26)	Required	Required	Required (workload)	Required	Required	Required	Required	Required	Required	Required
Recovery/DR/game-day (Ch.24)	Required	Required	Required (scenario)	Required	Required	Required	Required (named individual)	Required	Required	Required

Production verification (Ch.38)	Requ ired	Required	Requir ed (boun ded scope)	Requ ired	Required	Require d	Requir ed (name d individ ual)	Requi red	Requ ired	Require d (independ ent)
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30.R RACI / Control Ownership

Table 36. Chapter 30 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Verification)	Consulted	Informed
Confirm evidence retention/assurance	Accountable Authority Class (to be assigned)	Test Evidence Owner (to be assigned)	NCIE-010 owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [LEGAL/POLICY CONFIRMATION REQUIRED] (Blocking) HR16-30-1: Confirm evidence retention/assurance (Table 35; no retention period invented by this document).

Traceability: NCIE-001–015.

Test Execution Governance & Independence

Define executor/witness/reviewer/approver independence.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Specification only; no test in this document is executed, passed, or accepted by virtue of being specified here.

Upstream: NCIE-008 RACI/SoD.

31.0 Salient Things to Note

Executor ≠ approver; independence scales with risk/governance — the person running a test is never, by that act alone, also the person accepting its result.

31.1 Executor, Witness, Reviewer and Approver Roles

Executor ≠ approver — the person or automation running a test is never, by that act alone, also the person accepting its result. Witness and reviewer roles are added as risk increases, per NCIE-008's separation-of-duties model.

31.2 SoD, Rerun and Exception Handling

Independence scales with risk/governance — a low-risk unit test may run without a witness, while a security or recovery test requires an independent reviewer; any exception to the default SoD model is recorded and time-bound, never a silent standing practice.

31.R RACI / Control Ownership

Table 37. Chapter 31 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Verification)	Consulted	Informed
Confirm independence/sign-off roles	Accountable Authority Class (to be assigned)	Execution Governance Owner (to be assigned)	NCIE-008 owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR16-31-1: Confirm independence/sign-off roles.

Traceability: NCIE-008 RACI/SoD.

Acceptance, Sign-Off & Release-Readiness Matrix

Define acceptance/sign-off/release-readiness matrix.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Specification only; no test in this document is executed, passed, or accepted by virtue of being specified here.

Upstream: NCIE-008 acceptance; NCIE-015 go-live.

32.0 Salient Things to Note

TEST PASSED ≠ PRODUCTION ACCEPTANCE; blockers/residual defects remain visible — Table 38's Decision/Conditions column reads Pending throughout because no acceptance decision has actually been taken at the specification stage.

32.1 Capability, Tests, Results and Residual Risk

TEST PASSED ≠ PRODUCTION ACCEPTANCE — a capability's every relevant test passing is a precondition for acceptance consideration, never the acceptance decision itself. Blockers and residual defects remain visible in the acceptance record rather than hidden behind an aggregate status.

32.2 HR Blockers, Approver and Conditional Acceptance

Open Blocking Human Review items (Chapter 39) are explicit inputs to the acceptance decision; an approver may grant conditional acceptance with stated conditions, but the conditions and the approver's identity are recorded exactly as precisely as an unconditional acceptance would be.

32.3 Acceptance/Sign-Off Matrix (directive item 66)

The consolidated matrix below maps capability/risk class to required evidence, test status, residual defects/gaps, HR blockers, reviewer/approver authority and conditional/final acceptance, illustratively — actual entries populate as each capability's verification proceeds.

Table 38. Acceptance/Sign-Off Matrix (illustrative structure).

Capability/Risk Class	Required Evidence	Test Status	Residual Defects/Gap s	HR Blockers	Reviewer/Approver Authority	Decision/Conditions
Security-critical (e.g. Ch.19 authz)	Table 35 Security class evidence	NOT RUN (spec stage)	TBD upon execution	HR16-19-1 and related upstream	NCIE-008 accountable authority + independent security reviewer	Pending — no decision recorded
Recovery-critical (e.g. Ch.24 DR)	Table 35 Recovery/DR class evidence	NOT RUN (spec stage)	TBD upon execution	HR16-24-1 and related upstream	NCIE-008 accountable authority + independent recovery reviewer	Pending — no decision recorded
Domain-critical (e.g. Ch.21 modules)	Table 35 Integration/System class evidence	NOT RUN (spec stage)	TBD upon execution	HR16-21-1 and related upstream	NCIE-008 accountable authority	Pending — no decision recorded
Standard capability	Table 35 relevant class evidence	NOT RUN (spec stage)	TBD upon execution	As applicable	NCIE-008 accountable authority	Pending — no decision recorded

32.R RACI / Control Ownership

Table 39. Chapter 32 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Verification)	Consulted	Informed
Confirm final/conditional acceptance authority	Accountable Authority Class (to be assigned)	Acceptance Matrix Owner (to be assigned)	NCIE-008/015 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR16-32-1: Confirm final/conditional acceptance authority (Table 38).

Traceability: NCIE-008 acceptance; NCIE-015 go-live.

Coverage, Quality Metrics & Residual-Gap Reporting

Define coverage/quality/residual-gap reporting.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Specification only; no test in this document is executed, passed, or accepted by virtue of being specified here.

Upstream: NCIE-001–015.

33.0 Salient Things to Note

High coverage ≠ defect absence; untested/unverifiable requirements remain explicit — a residual gap is never omitted because reporting it would lower the coverage number.

33.1 Requirement/Test Coverage and Pass/Fail/Not-Run

Requirement coverage (fraction of RTM entries with a mapped test) and test coverage (fraction of mapped tests executed) are reported separately; high coverage ≠ defect absence — full coverage tells you every path was checked, not that every check found nothing.

33.2 Defect Trends, Flake and Residual Gaps

Defect trends and flake rate are reported over time, not as a single snapshot; untested or unverifiable requirements remain explicit residual gaps in the report, never silently omitted because they would lower the coverage number.

33.R RACI / Control Ownership

Table 40. Chapter 33 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Verification)	Consulted	Informed
Confirm quality metrics/thresholds	Accountable Authority Class (to be assigned)	Coverage Reporting Owner (to be assigned)	NCIE-001-015 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR16-33-1: Confirm quality metrics/thresholds (no numeric value invented by this document).

Traceability: NCIE-001–015.

Test Reproducibility, Flakiness & Result Integrity

Govern reproducibility, flaky tests and reruns.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Specification only; no test in this document is executed, passed, or accepted by virtue of being specified here.

Upstream: NCIE-001–015.

34.0 Salient Things to Note

Intermittent pass cannot silently become stable pass; reruns preserve evidence — a flaky test is investigated, never simply retried until green.

34.1 Seeds, Data, Config and Repeatability

Deterministic seeds, data snapshots and configuration are recorded per run so a result can be reproduced; environment drift (Chapter 7) between runs is itself tracked as a potential confound rather than ignored.

34.2 Flake Classification, Quarantine and Rerun Evidence

Intermittent pass cannot silently become stable pass — a test that sometimes fails is flagged flaky and quarantined per policy, investigated, and only returned to the blocking suite once its root cause is understood; every rerun preserves the original failing evidence rather than discarding it.

34.R RACI / Control Ownership

Table 41. Chapter 34 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Verification)	Consulted	Informed
Confirm flake/rerun policy	Accountable Authority Class (to be assigned)	Reproducibility Owner (to be assigned)	NCIE-001-015 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED] (Non-Blocking) HR16-34-1: Confirm flake/rerun policy; does not block the quarantine-before-blocking-suite discipline already defined in the interim.

Traceability: NCIE-001–015.

Verification of Human Review & Governance Controls

Verify implemented governance/approval/exception controls.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Specification only; no test in this document is executed, passed, or accepted by virtue of being specified here.

Upstream: NCIE-001–015.

35.0 Salient Things to Note

Governance-control testing does not resolve Human Review; it verifies implemented resolved rules — an expired approval must actually be rejected, not merely flagged.

35.1 Approval Binding, SoD and Recusal

Tests confirm approval-binding, separation-of-duties and recusal controls actually enforce what NCIE-008 specifies (e.g. an approver cannot also be the requester); governance-control testing does not resolve Human Review — it verifies that already-resolved rules are correctly implemented.

35.2 Expiry, Exception, Escalation and Break-Glass

Expiry, conditional-approval, exception, escalation and break-glass post-review controls are each tested for correct enforcement, including the negative case (an expired approval must actually be rejected, not merely flagged).

35.R RACI / Control Ownership

Table 42. Chapter 35 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Verification)	Consulted	Informed
Confirm governance-control test authority	Accountable Authority Class (to be assigned)	Governance-Control Test Owner (to be assigned)	NCIE-008 owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR16-35-1: Confirm governance-control test authority.
- Traceability: NCIE-001–015.

Cross-Specification End-to-End Scenarios

Define E2E scenarios spanning NCIE-001–015.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Specification only; no test in this document is executed, passed, or accepted by virtue of being specified here.

Upstream: NCIE-001–015.

36.0 Salient Things to Note

E2E success must preserve data→Evidence→analysis→Human decision→Tool/action→receipt→audit boundaries — collapsing any one boundary is a failed scenario, regardless of what the final screen showed.

36.1 Actors, Data, Context and Agent/Model/VPF

E2E scenarios trace a complete journey across actors, data, Memory/Context, Agent/model orchestration and VPF validation, exercising the full stack rather than any single specification's boundary alone.

36.2 Evidence, Approval, Tool, Audit and the Preserved Boundary Chain

E2E success must preserve data→Evidence→analysis→Human decision→Tool/action→receipt→audit boundaries — a scenario that appears to succeed end to end while collapsing any one of these boundaries (e.g. an analytical signal silently treated as Evidence) is a failed scenario, not a passed one, regardless of what the final screen showed.

36.R RACI / Control Ownership

Table 43. Chapter 36 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Verification)	Consulted	Informed
Confirm priority E2E scenarios	Accountable Authority Class (to be assigned)	E2E Scenario Owner (to be assigned)	NCIE-001-015 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR16-36-1: Confirm priority E2E scenarios.

Traceability: NCIE-001–015.

Test Automation in CI/CD & Continuous Verification

Integrate approved automated tests into CI/CD without self-authorizing release.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Specification only; no test in this document is executed, passed, or accepted by virtue of being specified here.

Upstream: NCIE-015 CI/CD; NCIE-004 test stack.

37.0 Salient Things to Note

Continuous automation does not self-authorize release — a green CI/CD pipeline remains an input to the Chapter 32 acceptance decision, never a substitute for it.

37.1 Triggers, Suites, Environments and Gates

Continuous-verification triggers, suite composition, target environments and gate placement (Chapters 6-7, NCIE-015 Ch.7) are specified; continuous automation does not self-authorize release — a green CI/CD pipeline remains an input to the Chapter 32 acceptance decision, never a substitute for it.

37.2 Artifacts, Quarantine and Release Evidence

CI/CD-produced artifacts (Chapter 30 evidence, Chapter 34 quarantine state) feed the acceptance matrix directly, so release evidence has an unbroken chain back to the actual pipeline run rather than a manually re-typed summary.

37.R RACI / Control Ownership

Table 44. Chapter 37 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Verification)	Consulted	Informed
Confirm blocking/advisory automated gates	Accountable Authority Class (to be assigned)	CI/CD Verification Owner (to be assigned)	NCIE-004/015 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED] (Blocking) HR16-37-1: Confirm blocking/advisory automated gates.

Traceability: NCIE-015 CI/CD; NCIE-004 test stack.

Production Verification, Smoke & Post-Deployment Validation

Define bounded production smoke/post-deployment verification.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Specification only; no test in this document is executed, passed, or accepted by virtue of being specified here.

Upstream: NCIE-001–015.

38.0 Salient Things to Note

Smoke pass ≠ full production acceptance; protect users/data and preserve authority — testing in production carries no implicit exemption from audit or privacy requirements.

38.1 Pre/Post Checks and Smoke/Synthetic Transactions

Bounded pre- and post-deployment checks and smoke/synthetic transactions verify the deployed system is minimally healthy; smoke pass ≠ full production acceptance — a smoke test's narrow scope is exactly why it runs fast, and exactly why it cannot substitute for full acceptance.

38.2 Rollback Triggers, Audit and Privacy in Production

Production verification tests carry a pre-defined rollback trigger, and — because they touch live systems — are scoped to protect real users and data, preserving audit and privacy requirements identically to any other production activity rather than granting testing an implicit exemption.

38.R RACI / Control Ownership

Table 45. Chapter 38 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Verification)	Consulted	Informed
Confirm allowed production tests/authority	Accountable Authority Class (to be assigned)	Production Verification Owner (to be assigned)	NCIE-001-015 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED] (Blocking) HR16-38-1: Confirm allowed production tests/authority.

Traceability: NCIE-001–015.

Human Review Register & Open Testing Decisions

Consolidate every unresolved testing decision exposed across Chapters 1-38 and Chapter 40 into one register: deduplicated, consistently classified, each stating precisely what it blocks. No item here manufactures closure of a decision the approved upstream chain, the approved NCIE-008–015 governance, security, persistence, domain-business and operations layers, or the approved NCIE-016 Skeletal v0.1 does not itself settle.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-008–015 Human Review registers, plus testing-specific unresolved items from Chapters 1-38.

39.0 Salient Things to Note

Do not invent acceptance thresholds, test tools, scenarios or authorities; deduplicate upstream decisions.

39.1 Purpose

This register exists so that no unresolved testing decision is silently settled by expansion prose, and so that no chapter's own Human Review Focus item is left outside the register it is meant to feed — a defect found and corrected in the companion NCIE-008 v1.0 document's initial build. Every item from Chapters 1-38 below was extracted programmatically from that chapter's own Human Review Focus section (§39.4). Chapter 39's own item (HR16-39-1) and Chapter 40's own item (HR16-40-1) are each defined once as a fixed value and rendered identically both here and in their own chapter's Human Review Focus bullet, so neither can be omitted from the register or drift from the bullet that also states it.

39.2 Summary

Table 46. Human Review Register summary (mechanically computed).

Classification	Count
ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED	5
INSTITUTIONAL CONFIRMATION REQUIRED	27
LEGAL/POLICY CONFIRMATION REQUIRED	1
RESOLVED BY UPSTREAM BASELINE	1
SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED	6
Total	40
— Blocking	38
— Non-Blocking	2

39.3 Complete Register

Table 47. Complete Human Review Register.

Review ID	Source Chapter	Question	Classification	Blocking Scope	Upstream Cross-Reference	Status
HR16-1-1	Chapter 1	Confirm the NCIE-016 formal title reconciliation (§0.1 — corrected to 'Testing, Verification & Acceptance	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm the NCIE-016 formal title reconciliation (§0.1 — corrected to 'Testing, Verification & Acceptance	N/A — distinct testing decision	Open

		Specification' against Master Controlled Register v2.0), and verification/sign-off authority (skeletal Preliminary Human Review Theme 1).		Specification' against Master Controlled Register v2.0), and verification/sign-off authority (skeletal Preliminary Human Review Theme 1).		
HR16-2-1	Chapter 2	Approve QA doctrine/independence (Table 4 state model).	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Approve QA doctrine/independence (Table 4 state model).	N/A — distinct testing decision	Open
HR16-3-1	Chapter 3	Confirm baseline-freeze/ambiguity escalation policy.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm baseline-freeze/ambiguity escalation policy.	N/A — distinct testing decision	Open
HR16-4-1	Chapter 4	Confirm RTM ownership/acceptance (Table 7 structure).	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm RTM ownership/acceptance (Table 7 structure).	N/A — distinct testing decision	Open
HR16-5-1	Chapter 5	Approve levels by risk.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Approve levels by risk.	N/A — distinct testing decision	Open
HR16-6-1	Chapter 6	Confirm prioritization/risk acceptance.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm prioritization/risk acceptance.	N/A — distinct testing decision	Open
HR16-7-1	Chapter 7	Confirm environment equivalence.	ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED (Blocking)	Confirm environment equivalence.	N/A — distinct testing decision	Open
HR16-8-1	Chapter 8	Confirm permitted production-derived data.	SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED (Blocking)	Confirm permitted production-derived data.	N/A — distinct testing decision	Open
HR16-9-1	Chapter 9	Confirm automation technology/coverage goals (no product mandated by this document).	ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED (Blocking)	Confirm automation technology/coverage goals (no product mandated by this document).	N/A — distinct testing decision	Open
HR16-10-1	Chapter 10	Confirm quality gates/coverage targets (no numeric value invented by this document).	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm quality gates/coverage targets (no numeric value invented by this document).	N/A — distinct testing decision	Open
HR16-11-1	Chapter 11	Confirm partner/certification acceptance.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm partner/certification acceptance.	N/A — distinct testing decision	Open
HR16-12-1	Chapter 12	Confirm migration/reconciliation acceptance.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm migration/reconciliation acceptance.	N/A — distinct testing	Open

					decision	
HR16-13-1	Chapter 13	Confirm Memory scenarios/acceptance.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm Memory scenarios/acceptance.	N/A — distinct testing decision	Open
HR16-14-1	Chapter 14	Confirm behavioral scenario corpus.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm behavioral scenario corpus.	N/A — distinct testing decision	Open
HR16-15-1	Chapter 15	Confirm Agent corpus/generated-code scope.	SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED (Blocking)	Confirm Agent corpus/generated-code scope.	N/A — distinct testing decision	Open
HR16-16-1	Chapter 16	Confirm benchmarks/evaluation sets/thresholds (no value invented by this document).	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm benchmarks/evaluation sets/thresholds (no value invented by this document).	N/A — distinct testing decision	Open
HR16-17-1	Chapter 17	Confirm high-risk Tool scenarios.	SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED (Blocking)	Confirm high-risk Tool scenarios.	N/A — distinct testing decision	Open
HR16-18-1	Chapter 18	Confirm VPF golden cases/authority.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm VPF golden cases/authority.	N/A — distinct testing decision	Open
HR16-19-1	Chapter 19	Confirm security-test independence/acceptance.	SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED (Blocking)	Confirm security-test independence/acceptance.	N/A — distinct testing decision	Open
HR16-20-1	Chapter 20	Confirm defensibility evidence/acceptance.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm defensibility evidence/acceptance.	N/A — distinct testing decision	Open
HR16-21-1	Chapter 21	Confirm domain scenarios/validators.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm domain scenarios/validators.	N/A — distinct testing decision	Open
HR16-22-1	Chapter 22	Confirm accessibility level/user groups.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm accessibility level/user groups.	N/A — distinct testing decision	Open
HR16-23-1	Chapter 23	Confirm operational test acceptance.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm operational test acceptance.	N/A — distinct testing decision	Open
HR16-24-1	Chapter 24	Confirm recovery scenarios/RPO-RTO acceptance.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm recovery scenarios/RPO-RTO acceptance.	N/A — distinct testing decision	Open
HR16	Chapter	Confirm workload	ENGINEERING/INFRASTRU	Confirm workload	N/A —	Open

-25-1	er 25	models/targets (no numeric value invented by this document).	CTURE DISCOVERY REQUIRED (Blocking)	models/targets (no numeric value invented by this document).	distinct testing decision	
HR16-26-1	Chapt er 26	Confirm fault-injection authority/safe environments.	SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED (Blocking)	Confirm fault-injection authority/safe environments.	N/A — distinct testing decision	Open
HR16-27-1	Chapt er 27	Confirm incident exercise/acceptance.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm incident exercise/acceptance.	N/A — distinct testing decision	Open
HR16-28-1	Chapt er 28	Confirm regression/release blocking policy.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm regression/release blocking policy.	N/A — distinct testing decision	Open
HR16-29-1	Chapt er 29	Confirm defect severity/closure authority (Table 33).	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm defect severity/closure authority (Table 33).	N/A — distinct testing decision	Open
HR16-30-1	Chapt er 30	Confirm evidence retention/assurance (Table 35; no retention period invented by this document).	LEGAL/POLICY CONFIRMATION REQUIRED (Blocking)	Confirm evidence retention/assurance (Table 35; no retention period invented by this document).	N/A — distinct testing decision	Open
HR16-31-1	Chapt er 31	Confirm independence/sign-off roles.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm independence/sign-off roles.	N/A — distinct testing decision	Open
HR16-32-1	Chapt er 32	Confirm final/conditional acceptance authority (Table 38).	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm final/conditional acceptance authority (Table 38).	N/A — distinct testing decision	Open
HR16-33-1	Chapt er 33	Confirm quality metrics/thresholds (no numeric value invented by this document).	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm quality metrics/thresholds (no numeric value invented by this document).	N/A — distinct testing decision	Open
HR16-34-1	Chapt er 34	Confirm flake/rerun policy; does not block the quarantine-before-blocking-suite discipline already defined in the interim.	ENGINEERING/INFRASTRU CTURE DISCOVERY REQUIRED (Non-Blocking)	Non-blocking — does not gate chapter-level verification activity	N/A — distinct testing decision	Open
HR16-35-1	Chapt er 35	Confirm governance-control test authority.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm governance-control test authority.	N/A — distinct testing decision	Open
HR16-36-1	Chapt er 36	Confirm priority E2E scenarios.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm priority E2E scenarios.	N/A — distinct testing	Open

					decision	
HR16-37-1	Chapter 37	Confirm blocking/advisory automated gates.	ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED (Blocking)	Confirm blocking/advisory automated gates.	N/A — distinct testing decision	Open
HR16-38-1	Chapter 38	Confirm allowed production tests/authority.	SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED (Blocking)	Confirm allowed production tests/authority.	N/A — distinct testing decision	Open
HR16-39-1	Chapter 39	Review this consolidated register and decide, for each Blocking item, whether it must be resolved before NCIE-016 v1.0 may be approved or may instead remain open under an interim design default.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Review this consolidated register and decide, for each Blocking item, whether it must be resolved before NCIE-016 v1.0 may be approved or may instead remain open under an interim design default.	N/A — distinct testing decision	Open
HR16-40-1	Chapter 40	Approve the NCIE-016 skeletal structure and authorize controlled full expansion — already satisfied by the approved NCIE-016 Skeletal v0.1 and the production directive that authorized this v1.0 expansion; preserved here as historical record, not a live open item.	RESOLVED BY UPSTREAM BASELINE (Non-Blocking)	Non-blocking — does not gate chapter-level verification activity	N/A — distinct testing decision	Resolved by upstream baseline

39.4 Mechanical Verification

38 of these 40 items were recomputed programmatically from the Human Review Focus bullet text of Chapters 1-38 rather than assembled by hand. The remaining 2 items are HR16-39-1 (this chapter's own item) and HR16-40-1 (Chapter 40's own item), each defined once and rendered identically in both this table and their own chapter's Human Review Focus bullet. 0 duplicate Review IDs; 0 orphan Human Review items — every Human Review ID appearing anywhere in this specification is reconciled into this register.

39.5 Upstream Deduplication (Directive Item 70)

Every NCIE-016 Human Review item was individually checked against the existing NCIE-008–015 Human Review registers. No NCIE-016 item was found to restate an identical authority, threshold, environment, security, UX or operations question — each concerns a distinct testing decision (test-tool selection, coverage/threshold value, scenario corpus, or acceptance authority) that those upstream documents raise the need for but do not themselves resolve. No cross-reference deduplication was therefore required for this edition.

39.6 Blocking Scope Discipline

Each Blocking item states exactly what it blocks — a specific tool selection, numeric threshold, scenario corpus or acceptance authority — rather than implying the whole of NCIE-016 or the whole approved upstream chain is

blocked. Non-Blocking items are explicitly proposed defaults or confirmations that do not gate the corresponding chapter's verification activity in the interim.

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR16-39-1: Review this consolidated register and decide, for each Blocking item, whether it must be resolved before NCIE-016 v1.0 may be approved or may instead remain open under an interim design default.

Traceability: All chapters above (mechanically reconciled register); NCIE-008–015 Human Review registers.

Traceability, Acceptance & Final Verification Handover

Close traceability to the approved NCIE-001-015 foundation and the approved NCIE-016 Skeletal v0.1; define acceptance criteria and the final controlled handover.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Specification only; no test in this document is executed, passed, or accepted by virtue of being specified here.

Upstream: Approved NCIE-001-015 foundation.

40.0 Salient Things to Note

Skeleton approval authorizes expansion only; testing architecture approval is not executed verification — §40.4's acceptance gate states this explicitly, and no test in this specification has been executed.

40.1 Traceability Matrix

Every chapter of this document traces to an approved upstream authority or to the approved NCIE-016 Skeletal v0.1 itself. No chapter states a verification requirement without an identifiable source.

Table 48. Full traceability matrix.

NCIE-016 Chapter	Primary Upstream Authority	Relationship
Chapter 1	NCIE-001-015	Mandate/authority chain/verification boundary
Chapter 2	NCIE-001-015	Quality engineering & verification doctrine
Chapter 3	NCIE-001-015	Requirements baseline & testable acceptance extraction
Chapter 4	NCIE-004 comprehensive verification programme	RTM architecture
Chapter 5	NCIE-001-015	Test levels & verification taxonomy
Chapter 6	NCIE-001-015	Risk-based test prioritization
Chapter 7	NCIE-001-015	Test environment & configuration control
Chapter 8	NCIE-001-015	Test data, synthetic data & privacy
Chapter 9	NCIE-001-015	Test automation architecture
Chapter 10	NCIE-001-015	Static analysis, code quality & unit verification
Chapter 11	NCIE-011 Ch.30	API, contract & integration testing
Chapter 12	NCIE-003; NCIE-014 Ch.32	Data model, database & migration testing
Chapter 13	NCIE-006 Ch.34; NCIE-009	Memory, Context, search & retrieval testing
Chapter 14	NCIE-005; NCIE-012	ARGUS behavioral & conversational testing
Chapter 15	NCIE-007 Ch.32; NCIE-009	Agent Factory, runtime & multi-Agent testing
Chapter 16	NCIE-007/009/013	Model Gateway & AI evaluation
Chapter 17	NCIE-007/011/009	Tool Gateway & consequential-action testing

Chapter 18	NCIE-003/007/009	VPF validation system testing
Chapter 19	NCIE-009 Ch.26	Security, IAM & Zero-Trust testing
Chapter 20	NCIE-010 Ch.27	Evidence, provenance, audit & defensibility testing
Chapter 21	NCIE-013 Ch.32	Functional module & domain scenario validation
Chapter 22	NCIE-012 Ch.35	UX, accessibility & representative-user validation
Chapter 23	NCIE-015 Ch.34	DevSecOps, deployment & pipeline testing
Chapter 24	NCIE-014/015/009/010	Backup, restore, DR & game-day validation
Chapter 25	NCIE-002 NFRs; NCIE-014/015	Performance, load, scalability & capacity testing
Chapter 26	NCIE-004 resilience; NCIE-015	Resilience, fault injection & chaos validation
Chapter 27	NCIE-015 Ch.20-24; NCIE-010	Observability, alerting & incident-response verification
Chapter 28	NCIE-001-015	Regression, compatibility & change-impact testing
Chapter 29	NCIE-001-015	Defect, triage, root-cause & corrective-action governance
Chapter 30	NCIE-001-015	Test evidence, results & verification artifact model
Chapter 31	NCIE-008 RACI/SoD	Test execution governance & independence
Chapter 32	NCIE-008 acceptance; NCIE-015 go-live	Acceptance, sign-off & release-readiness matrix
Chapter 33	NCIE-001-015	Coverage, quality metrics & residual-gap reporting
Chapter 34	NCIE-001-015	Test reproducibility, flakiness & result integrity
Chapter 35	NCIE-001-015	Verification of Human Review & governance controls
Chapter 36	NCIE-001-015	Cross-specification end-to-end scenarios
Chapter 37	NCIE-015 CI/CD; NCIE-004 test stack	Test automation in CI/CD & continuous verification
Chapter 38	NCIE-001-015	Production verification, smoke & post-deployment validation
Chapter 39	All chapters above; NCIE-008-015 Human Review registers	Consolidated Human Review Register
Chapter 40	Approved NCIE-001-015; NCIE-016 Skeletal v0.1	Traceability/acceptance/final handover

40.2 Chapter-Count and Salient-Note Mechanical Verification

A programmatic scan of this document confirms exactly 40 sequential 'NCIE-016 — Chapter N Expansion' labels, N = 1..40, each present exactly once, in order, with no gap, duplicate or unintended renaming (directive item 85). A second scan confirms an explicit '<N>.0 Salient Things to Note' heading present for all 40 chapters (directive item 86), rerun independently against this final document rather than assumed from the build script.

40.3 Skeleton-to-Full Fidelity Verification

Every approved Chapter Purpose, Salient Things to Note, Principal Component, Upstream Dependency/Authority and Human Review Focus from the approved NCIE-016 Skeletal v0.1 is substantively represented in the corresponding full chapter above (directive item 84); none is omitted, and none is conceptually inverted.

40.4 Acceptance Criteria and the Unresolved-Item Gate

Full approval of NCIE-016 v1.0 requires: (a) 40/40 chapter integrity confirmed; (b) 40/40 explicit Salient Things to Note subsections confirmed; (c) skeleton-to-full fidelity confirmed; (d) every Blocking Human Review item in Chapter 39 either resolved by an institutional decision or explicitly and knowingly deferred by the approving authority (including the formal-title reconciliation, §0.1, HR16-1-1); (e) confirmation that no test anywhere in this document is falsely marked executed, passed, validated, or accepted; and (f) three-artifact consistency (this document, the Change Register and the Verification Report) confirmed. Approval of the v0.1 skeleton authorized this expansion only — it did not itself approve any tool selection, numeric threshold, acceptance authority or scenario-corpus assignment proposed above, and it did not execute a single test. Testing-architecture approval under this chapter does not prove requirement correctness, implementation correctness, security assurance, recovery capability, performance capacity or production readiness — those require actual, executed, evidence-backed verification against the acceptance matrix (Chapter 32), which remains entirely downstream of this document.

40.5 Final Verification Handover Register

NCIE-016 does not preempt actual test execution. The table below states what is handed over and to whom.

Table 49. Final verification handover register.

Downstream Consumer

Engineering/execution teams

NCIE-008 institutional acceptance authority

Subsequent controlled documents

What NCIE-016 Hands Over

The complete specification-stage RTM (Table 7), test-evidence model (Table 35) and acceptance matrix structure (Table 38) as the basis for actual test-case authoring and execution; this document supplies the programme, not the executed results.

The consolidated Human Review Register (Chapter 39) and acceptance/sign-off matrix structure as the basis for eventual, evidence-backed production acceptance decisions once tests are actually executed.

The complete set of open testing/verification Human Review items (Chapter 39) as dependencies for any later controlled document that consumes NCIE-016's verification architecture.

40.6 Readiness and Final Disposition

NCIE-016 Testing, Verification & Acceptance Specification v1.0 — Full Expansion of Approved Skeletal v0.1 — is READY FOR HUMAN REVIEW. It is NOT marked Approved. The approved v0.1 skeleton remains historically preserved as the expansion architecture; every proposed tool selection, numeric threshold, acceptance authority and scenario-corpus assignment deferred in Chapters 1-38 remains a proposal pending the Chapter 39 Human Review Register. No test described anywhere in this specification has been executed, passed, validated, or accepted — this document is a verification programme, not a verification result.

Human Review Focus

- [RESOLVED BY UPSTREAM BASELINE] (Non-Blocking) HR16-40-1: Approve the NCIE-016 skeletal structure and authorize controlled full expansion — already satisfied by the approved NCIE-016 Skeletal v0.1 and the production directive that authorized this v1.0 expansion; preserved here as historical record, not a live open item.

Traceability: NCIE-001-015 approved foundation; approved NCIE-016 Skeletal v0.1.